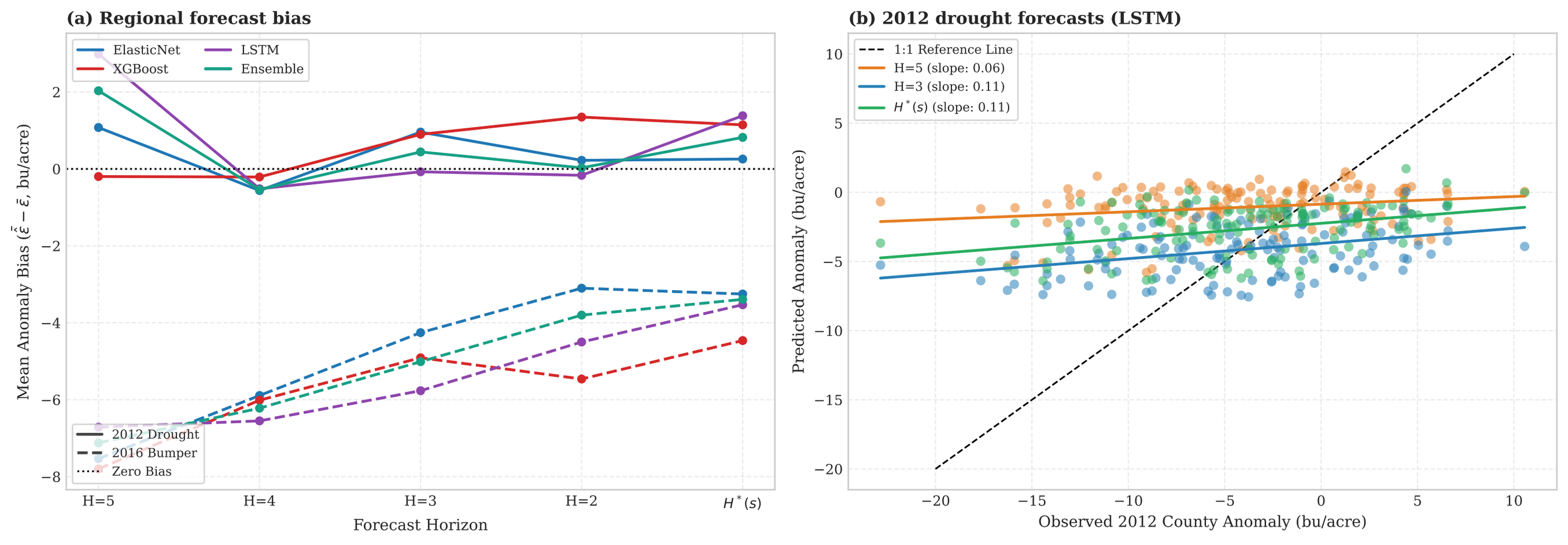
(a) Regional forecast bias
ElasticNet
XGBoost
LSTM
Ensemble
2012 Drought
2016 Bumper
Zero Bias
Mean Anomaly Bias ($\bar{\hat{\varepsilon}} - \bar{\varepsilon}$, bu/acre)
2
0
−2
−4
−6
−8
H=5
H=4
H=3
H=2
$H^*(s)$
Forecast Horizon
(b) 2012 drought forecasts (LSTM)
1:1 Reference Line
H=5 (slope: 0.06)
H=3 (slope: 0.11)
$H^*(s)$ (slope: 0.11)
Predicted Anomaly (bu/acre)
10
5
0
−5
−10
−15
−20
−20
−15
−10
−5
0
5
10
Observed 2012 County Anomaly (bu/acre)